Math 526, S12 Name: Answer

Quiz 4 Section: SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (6 points) Given

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 4 & 5 \\ 0 & 4 & 0 \end{bmatrix}.$$

Find $\mathbf{A}^{-1}$ using the Gaussian-Jordan Elimination.

$$[A\ I] = \left[\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 2 & 4 & 5 & 0 & 1 & 0 \\ 0 & 4 & 0 & 0 & 0 & 1 \end{array}\right]$$

$$= \left[\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 2 & 1 & -2 & 1 & 0 \\ 0 & 4 & 0 & 0 & 0 & 1 \end{array}\right] \leftarrow \text{subtract } 2\cdot\text{Row1 from Row2}$$

$$= \left[\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 2 & 1 & -2 & 1 & 0 \\ 0 & 0 & -2 & 4 & -2 & 1 \end{array}\right] \leftarrow \text{subtract } 2\cdot\text{Row2 from Row3}$$

$$= \left[\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 & 1/2 \\ 0 & 0 & -2 & 4 & -2 & 1 \end{array}\right] \leftarrow \text{add } (\tfrac{1}{2})\text{Row3 into Row2}$$

$$= \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 5 & -2 & 1 \\ 0 & 2 & 0 & 0 & 0 & 1/2 \\ 0 & 0 & -2 & 4 & -2 & 1 \end{array}\right] \leftarrow \text{add Row3 into Row1}$$

$$= \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 5 & -2 & 3/4 \\ 0 & 2 & 0 & 0 & 0 & 1/2 \\ 0 & 0 & -2 & 4 & -2 & 1 \end{array}\right] \leftarrow \text{subtract } (\tfrac{1}{2})\cdot\text{Row2 from Row1}$$

$$= \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 5 & -2 & 3/4 \\ 0 & 1 & 0 & 0 & 0 & 1/4 \\ 0 & 0 & 1 & -2 & 1 & -1/2 \end{array}\right] \leftarrow \text{scale the diagonal to 1}$$

$$\Rightarrow A^{-1} = \begin{bmatrix} 5 & -2 & 3/4 \\ 0 & 0 & 1/4 \\ -2 & 1 & -1/2 \end{bmatrix}$$

Problem 2. (4 points) Let $\mathbf{A} = \begin{bmatrix} 3 & 3 & 4 \\ 1 & -1 & 3 \\ -2 & 4 & -1 \end{bmatrix}$. Find the **LU** factorization of **A**.

$$A = \begin{bmatrix} 3 & 3 & 4 \\ 1 & -1 & 3 \\ -2 & 4 & -1 \end{bmatrix} \xrightarrow[\ell_{21}=\frac{1}{3}]{E_{21}} \begin{bmatrix} 3 & 3 & 4 \\ 0 & -2 & 5/3 \\ -2 & 4 & -1 \end{bmatrix} \xrightarrow[\ell_{31}=-2/3]{E_{31}} \begin{bmatrix} 3 & 3 & 4 \\ 0 & -2 & 5/3 \\ 0 & 6 & 5/3 \end{bmatrix}$$

$$\xrightarrow[\ell_{32}=-3]{E_{32}} \begin{bmatrix} 3 & 3 & 4 \\ 0 & -2 & 5/3 \\ 0 & 0 & 20/3 \end{bmatrix}$$

Thus

$A = LU$ with

$$L = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{3} & 1 & 0 \\ -\frac{2}{3} & -3 & 1 \end{bmatrix}, \quad U = \begin{bmatrix} 3 & 3 & 4 \\ 0 & -2 & \frac{5}{3} \\ 0 & 0 & \frac{20}{3} \end{bmatrix}$$